UNIT 2 · TOPICS 2.1–2.2 · TWO CATEGORICAL VARIABLES

Two Categorical Variables: Tables, Graphs & Association

Two-way (contingency) tables, joint / marginal / conditional relative frequencies, side-by-side and segmented bar charts, and the reasoning that turns a difference in conditional distributions into evidence of association — without claiming causation.

1. Two-Way Tables and the Three Relative Frequencies

A two-way table, also called a contingency table, summarises two categorical variables measured on the same individuals. Cell entries may be **counts** (frequencies) or **proportions** (relative frequencies). Always ask *"out of whom?"* before dividing — the denominator is the whole exam trap in this topic.

JOINT, MARGINAL AND CONDITIONAL RELATIVE FREQUENCY

joint relative frequency = cell / table total
marginal relative frequency = row total / table total (or column total / table total)
conditional relative frequency = cell / that row's total (or that column's total)

A **marginal** distribution describes one variable alone (the row/column totals in the margins). A **conditional** distribution describes one variable *within a single level* of the other — the one used to compare groups.

WORKED EXAMPLE	PASSED	FAILED	ROW TOTAL
Tutoring	42	18	60
No tutoring	30	30	60
Column total	72	48	**120**

- **Joint:** P(Tutoring and Passed) = 42/120 = 0.35.
- **Marginal:** P(Tutoring) = 60/120 = 0.50; P(Passed) = 72/120 = 0.60.
- **Conditional:** P(Passed | Tutoring) = 42/60 = 0.70 versus P(Passed | No tutoring) = 30/60 = 0.50.

2. Graphical Displays and Deciding Whether Variables Are Associated

Side-by-side bar charts, **segmented (stacked) bar charts** and **mosaic plots** all display the distribution of one categorical variable at each level of the other. Segmented bars must be scaled to 100% per group, otherwise unequal group sizes make the picture misleading.

ASSOCIATION RULE

conditional distributions differ across groups → the variables are associated
conditional distributions are (nearly) the same → no evidence of association

In the worked table, 70% vs 50% is a clear difference, so tutoring status and passing are **associated** in these data. State the direction with the numbers, in context.

ASSOCIATION IS NOT CAUSATION

An association in an observational two-way table does **not** justify "tutoring caused the pass rate". Only a randomized experiment that assigns the explanatory variable supports a causal conclusion; otherwise name a plausible confounding variable.

3. Interpretation Templates and Exam Traps

EXAM-SAFE WORDING

- "Among students who received tutoring, $42/60 = 70\%$ passed, compared with $30/60 = 50\%$ among those who did not. Because these conditional relative frequencies differ, the data suggest an association between tutoring and passing."
- "The marginal distribution of passing is $72/120 = 60\%$ passed and $48/120 = 40\%$ failed, ignoring tutoring status."

TRAPS THE EXAM ACTUALLY TESTS

- **Wrong denominator:** $42/120$ is a joint relative frequency, not $P(\text{Passed} \mid \text{Tutoring})$. Conditional claims divide by the row/column total.
- **Comparing raw counts** when group sizes differ — always convert to relative frequencies first.
- **Describing only one group.** A comparison needs at least two conditional distributions.
- **"No association" is not proof of independence**; say "the data do not show a difference".
- **Association ≠ causation** unless the data come from a randomized experiment.